MLOps – Assignment 01

Apache Pig Based Product Review Analysis Using Hadoop

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Objective

The objective of this assignment is to use Apache Pig on a Hadoop environment to process an e-commerce product review dataset and calculate the average rating for each product category.

Main objectives:

- Start and verify Hadoop HDFS services.
- Start and verify Hadoop YARN services.
- Store the review dataset in HDFS.
- Load the dataset using Apache Pig.
- Remove invalid ratings.
- Group valid reviews by product category.
- Calculate the average rating for each category.
- Store and verify the processed result in HDFS.

Problem Statement

An e-commerce dataset contains customer reviews for different product categories such as Home, Books, Clothing and Electronics. Some records contain invalid ratings. A null rating or a rating below 1.0 is treated as invalid. Apache Pig is used to remove these records, group the remaining reviews by category, and calculate the average rating for each category.

Technologies Used

Technology	Purpose
Apache Hadoop	Distributed data processing framework
HDFS	Storage of input dataset and output
YARN	Resource and job management
Apache Pig 0.17.0	Data processing using Pig Latin
Pig Latin	Data-flow scripting language
Linux / Ubuntu	Operating environment
MapReduce	Execution engine

Hadoop Environment

Before executing the Pig program, the Hadoop services were started. The **jps** command was used to verify the running Hadoop processes, including NameNode, DataNode, SecondaryNameNode, ResourceManager and NodeManager.

```
start-dfs.sh
jps
start-yarn.sh
jps
```

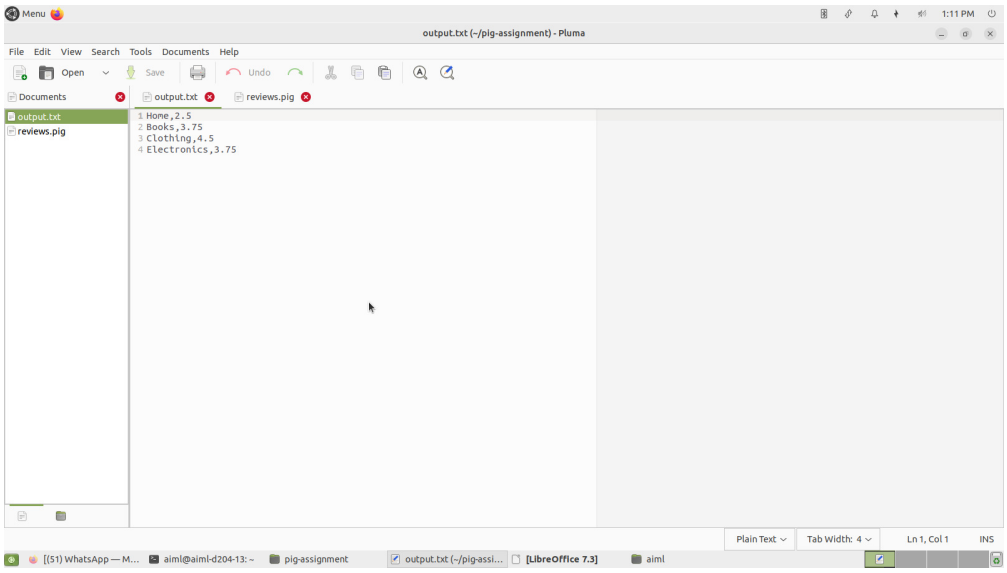


Figure 1: Hadoop environment / service verification

Apache Pig Environment

Apache Pig was used to process the review dataset using Pig Latin. The Pig installation was verified and the program was executed in MapReduce mode so that Hadoop handled the processing job.

Apache Pig version: 0.17.0

Input Dataset

The input file used for the experiment is reviews.csv. It contains three fields: Review ID, Category and Rating.

Review ID	Category	Rating
1	Electronics	4.5
2	Electronics	3.0
3	Electronics	0.5
4	Clothing	5.0
5	Clothing	4.0
6	Clothing	0.0
7	Books	4.0
8	Books	3.5
9	Books	
10	Home	2.5

Storing the Dataset in HDFS

The input dataset was stored in HDFS under the directory /user/aiml/reviews/. The file was then verified and displayed from HDFS.

```
hdfs dfs -mkdir -p /user/aiml/reviews
hdfs dfs -put -f reviews.csv /user/aiml/reviews/
hdfs dfs -ls /user/aiml/reviews
hdfs dfs -cat /user/aiml/reviews/reviews.csv
```

The HDFS input location is: **/user/aiml/reviews/reviews.csv**

Apache Pig Script

The Pig program loads the CSV file, filters invalid ratings, groups valid records by category, calculates the average rating and stores the result in HDFS.

```
reviews = LOAD '/user/aiml/reviews/reviews.csv'
          USING PigStorage(',')
          AS (review_id:int, category:chararray, rating:double);

valid_reviews = FILTER reviews
                BY rating IS NOT NULL AND rating >= 1.0;

grouped_reviews = GROUP valid_reviews BY category;

average_ratings = FOREACH grouped_reviews
                  GENERATE group AS category,
                  AVG(valid_reviews.rating) AS average_rating;

STORE average_ratings
INTO '/user/aiml/review_average_output'
USING PigStorage(',');
```

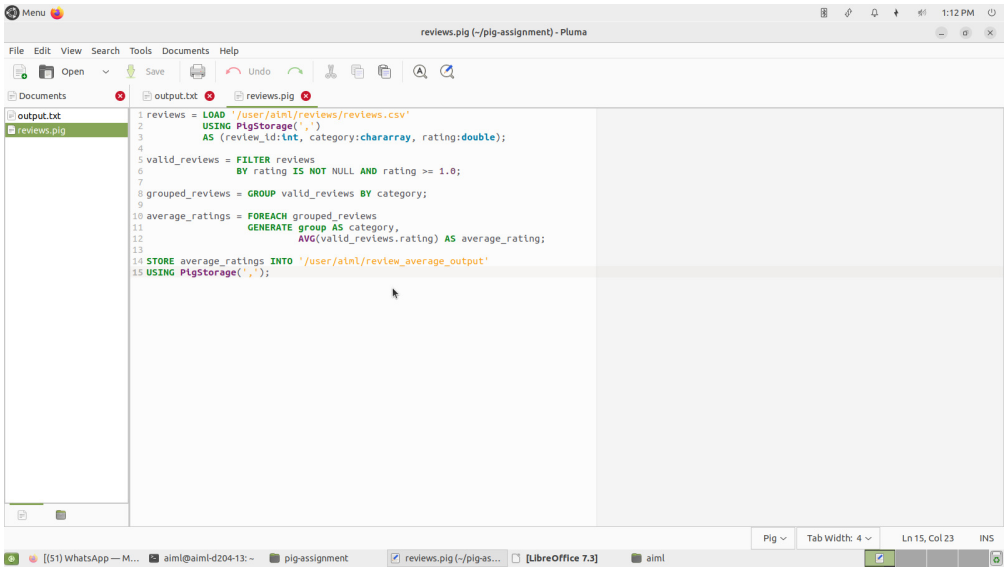


Figure 2: Pig Latin script used for the analysis

Explanation of the Pig Program

LOAD: The LOAD operation reads the reviews.csv dataset from HDFS. PigStorage(',') specifies that the input fields are separated by commas. The schema defines the review ID, category and rating data types.

```
reviews = LOAD '/user/aiml/reviews/reviews.csv'
USING PigStorage(',')
AS (review_id:int, category:chararray, rating:double);
```

FILTER: The filtering operation removes invalid records. The condition **rating IS NOT NULL AND rating >= 1.0** ensures that missing ratings and ratings below 1.0 are excluded.

```
valid_reviews = FILTER reviews
BY rating IS NOT NULL AND rating >= 1.0;
```

GROUP: The valid records are grouped according to their product category. This creates one group for each category.

```
grouped_reviews = GROUP valid_reviews BY category;
```

FOREACH and AVG: Each category group is processed and the AVG() function calculates its average rating.

```
average_ratings = FOREACH grouped_reviews
GENERATE group AS category,
AVG(valid_reviews.rating) AS average_rating;
```

STORE: The calculated category-wise averages are saved back into HDFS.

```
STORE average_ratings
INTO '/user/aiml/review_average_output'
USING PigStorage(',');
```

Execution

The Pig script was executed in MapReduce mode using the following command:

```
pig -x mapreduce ~/pig-assignment/reviews.pig
```

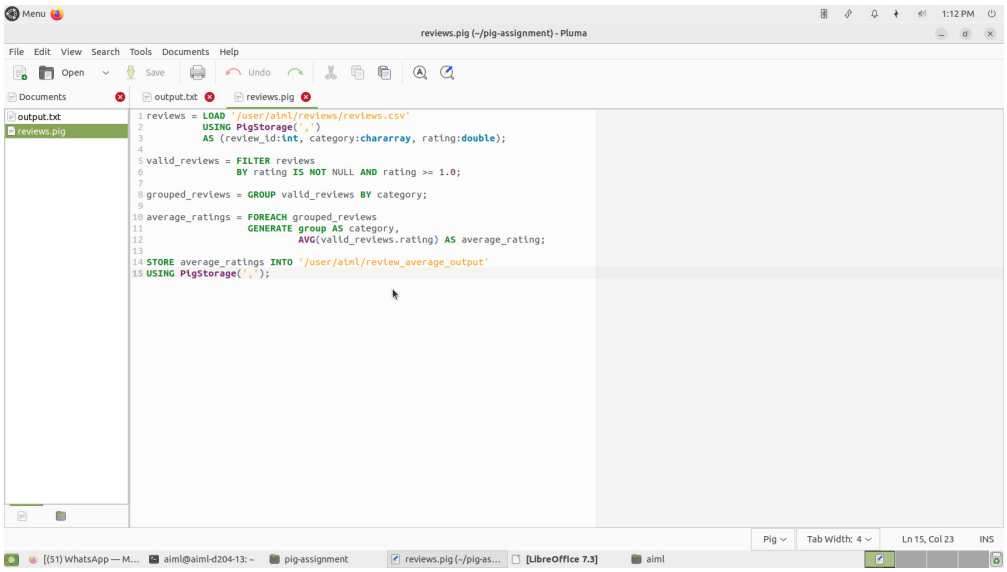


Figure 3: Execution of the Pig program in MapReduce mode

Output Verification

After successful execution, the generated result was stored in the HDFS directory **/user/aiml/review_average_output**. The output file was verified using the Hadoop command line.

```
hdfs dfs -ls /user/aiml/review_average_output
hdfs dfs -cat /user/aiml/review_average_output/part-r-00000
```

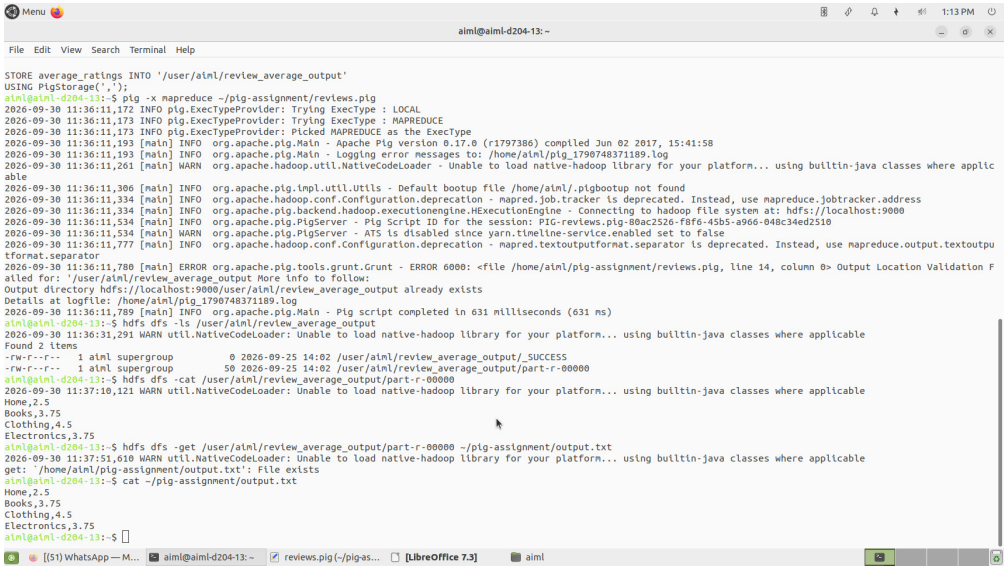


Figure 4: Final category-wise average ratings

Final Output

Category	Average Rating
Home	2.5
Books	3.75
Clothing	4.5
Electronics	3.75

The output confirms that only valid ratings were considered for the average calculation. Ratings below 1.0 and missing ratings were excluded before grouping and aggregation.

Result Analysis

Home: The Home category contains one valid rating, 2.5. Therefore, the average is 2.5.

```
Home average = 2.5
```

Books: The valid ratings are 4.0 and 3.5. The missing rating is excluded.

```
(4.0 + 3.5) / 2
= 7.5 / 2
= 3.75
```

Therefore, Books = 3.75.

Clothing: The valid ratings are 5.0 and 4.0. The 0.0 rating is removed because it is below 1.0.

```
(5.0 + 4.0) / 2
= 9.0 / 2
= 4.5
```

Therefore, Clothing = 4.5.

Electronics: The valid ratings are 4.5 and 3.0. The 0.5 rating is removed because it is below 1.0.

```
(4.5 + 3.0) / 2
= 7.5 / 2
= 3.75
```

Therefore, Electronics = 3.75.

Complete Processing Flow

Step	Operation	Description
1	reviews.csv	Input dataset
2	Store in HDFS	Distributed storage
3	LOAD	Read data using Pig
4	FILTER	Remove NULL and rating < 1.0
5	GROUP BY category	Create category groups
6	AVG()	Calculate category averages
7	STORE	Save result in HDFS
8	Verify output	Display part-r-00000

Important Commands

```
start-dfs.sh
jps
start-yarn.sh
hdfs dfs -cat /user/aiml/reviews/reviews.csv
pig -x mapreduce ~/pig-assignment/reviews.pig
hdfs dfs -ls /user/aiml/review_average_output
hdfs dfs -cat /user/aiml/review_average_output/part-r-00000
```

Role of Hadoop and Pig

Hadoop HDFS: Provides storage for the input review dataset and the generated output.

YARN: Manages the resources required for Hadoop jobs.

Apache Pig: Provides the Pig Latin data-flow language used for loading, filtering, grouping and aggregation.

MapReduce: Executes the Pig processing job using the Hadoop environment when MapReduce mode is selected.

Outcome

The Apache Pig program was successfully executed in MapReduce mode on the Hadoop environment. The review dataset was loaded from HDFS, invalid records containing ratings below 1.0 or missing ratings were filtered out, and the remaining records were grouped according to product category.

The average rating for each category was calculated using the AVG() function. The final category-wise results were successfully stored in HDFS and verified from the generated output file.

Conclusion

This assignment provided practical experience in using Apache Hadoop, HDFS, YARN and Apache Pig for data processing. The experiment demonstrated how a CSV dataset can be stored in HDFS and processed using Pig Latin.

The operations LOAD, FILTER, GROUP, FOREACH, AVG() and STORE were used to transform the raw review data into meaningful category-wise average ratings. The final output confirms that the required product review analysis was completed successfully.

Final Result Summary

Category	Average
Home	2.5
Books	3.75
Clothing	4.5
Electronics	3.75

— End of Report —